A Rigorous Proof Framework for Lower Bounds on the Autoconvolution Constant C_{1a}

Via the Cloninger–Steinerberger Branch-and-Prune Algorithm

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Abstract

We present a self-contained proof that the autoconvolution constant C_{1a} — the infimum of $\|f * f\|_{L^\infty}/\left(\int f\right)^2$ over nonnegative functions supported on $[-\frac{1}{4}, \frac{1}{4}]$ — satisfies the lower bound $C_{1a} \geq \frac{7}{5} = 1.4$, certifiable by exhaustive computation. Our starting point is the framework of Cloninger and Steinerberger [CS17], which we extend with a multiscale cascade and tighter pruning. The paper is organized around the natural sequence of reductions: from continuous functions to bin averages (Section 2), from the continuous simplex to a finite lattice (Section 3), and finally to an exhaustive search made tractable by pruning (Sections 4–6). Section 8 addresses the invariants that keep the computation honest. In the accompanying Lean formalization, the analytic reduction is proved directly and the external computation is packaged as a single terminal-grid certificate. Throughout, we emphasize intuition alongside formal statements.

Keywords. Autoconvolution, Sidon sets, branch-and-prune, computer-assisted proof, Lean formalization.

1. Introduction

1.1. The Constant C_{1a}

Consider the class of nonnegative, integrable functions supported on the interval $[-\frac{1}{4}, \frac{1}{4}]$:

$$\mathcal{F} \;=\; \left\{ f : \mathbb{R} \to \mathbb{R}_{\geq 0} \;\middle|\; \operatorname{supp}(f) \subseteq \left[-\tfrac{1}{4}, \tfrac{1}{4}\right],\; \int f > 0 \right\}.$$

The *autoconvolution* of f is

$$(f * f)(t) \;=\; \int_{-\infty}^{\infty} f(x)\, f(t-x)\, dx.$$

Since $\operatorname{supp}(f) \subseteq [-\frac{1}{4}, \frac{1}{4}]$, the autoconvolution is supported on $[-\frac{1}{2}, \frac{1}{2}]$ and is itself nonnegative.

A natural question is: how small can the peak of $f * f$ be, relative to the total mass of f? This is precisely what C_{1a} measures.

Definition 1.1. The *autoconvolution constant* is

$$C_{1a} \;=\; \inf_{f \in \mathcal{F}} \frac{\max\limits_{|t| \leq 1/2} (f * f)(t)}{\left(\int f\right)^2}.$$

Bound	Value	Reference
Published upper bound	≤ 1.50992	Matolcsi & Vinuesa [MV10]
Published lower bound	≥ 1.28	Cloninger & Steinerberger [CS17]
Lower bound	≥ 1.4	This work

Table 1: Published bounds on C_{1a} in the normalization used here.

Remark 1.2 (Open versus closed support)**.** The Lean formalization uses the open support condition $\mathrm{supp}(f) \subseteq (-\frac{1}{4}, \frac{1}{4})$. For the present problem, this is equivalent to the closed-interval formulation above: changing a function on the boundary points $\pm\frac{1}{4}$ does not affect its integral or autoconvolution ratio.

By homogeneity, we may normalize $\int f = 1$ without loss of generality, so that $C_{1a} = \inf_f \|f * f\|_{L^\infty}$ over nonnegative f with $\mathrm{supp}(f) \subseteq [-\frac{1}{4}, \frac{1}{4}]$ and $\int f = 1$.

Why this constant matters. C_{1a} is connected to the asymptotic theory of Sidon and generalized Sidon sets in additive combinatorics [ET41, MO06, CRV10, MO09, CS17]. A *Sidon set* is a set of integers where all pairwise sums are distinct; equivalently, the "representation function" of the sumset has values at most 2 everywhere except at doubled elements. The constant C_{1a} governs how efficiently mass can be distributed to minimize the peak of the autoconvolution — the continuous analog of controlling pairwise-sum collisions.

1.2. Known Bounds

The *upper bound* comes from exhibiting explicit functions whose autoconvolution peak is small. This is a constructive task: find a good f and compute its peak.

The *lower bound* is fundamentally harder: one must show that *no* function can achieve a peak below the threshold. This requires exhaustive elimination of all possibilities, which is the subject of this document.

For the longer historical chain of published improvements, together with announcement-level updates beyond the journal literature, see the optimization-constants repository maintained by Tao [Tao]. The published lower bounds of Martin–O'Bryant [MO07, MO09], Matolcsi–Vinuesa [MV10], and Cloninger–Steinerberger [CS17] give the progression

$$1 \;<\; 1.182778 \;<\; 1.262 \;<\; 1.2748 \;<\; 1.28$$

in the present normalization, while the older upper bound $\pi/2$ comes from Schinzel–Schmidt [SS02].

1.3. Proof Strategy at a High Level

The proof proceeds in three conceptual steps:

(i) **Reduce** the infinite-dimensional optimization (over all nonnegative functions) to a finite-dimensional one (over vectors of bin averages), with a quantified approximation error.

(ii) **Discretize** the finite-dimensional simplex to a lattice, with a quantified rounding error.

(iii) **Enumerate** every lattice point and verify that each one has autoconvolution peak exceeding the target threshold. If all are eliminated, the bound is proven.

Step (iii) is made tractable by pruning strategies (asymmetry, windowed bounds) and a multiscale cascade (coarse-to-fine refinement).

2. From Functions to Bin Averages

Our first order of business is to replace the infinite-dimensional optimization — over all nonnegative functions — with something finite-dimensional and computable.

2.1. The Partition

Fix a positive integer n (the "half-bin count"). Partition $(-\frac{1}{4}, \frac{1}{4})$ into $d = 2n$ equal bins:

$$I_j \;=\; \left(\frac{j}{4n},\ \frac{j+1}{4n}\right), \qquad j = -n, -n+1, \ldots, n-1.$$

Each bin has width $\frac{1}{4n}$.

Remark 2.1. We use $d = 2n$ bins total, indexed from $-n$ to $n-1$. In the implementation, we re-index these as $0, 1, \ldots, d-1$ (i.e., $j \mapsto j+n$). The mathematics is the same either way.

In the Lean development, the bins are taken to be half-open intervals

$$\left[-\tfrac{1}{4} + \frac{i}{4n},\ -\tfrac{1}{4} + \frac{i+1}{4n}\right),$$

so that they form a genuine partition. This change affects neither the bin integrals nor the convolution bounds, since the endpoints form a null set.

Figure 1 illustrates the support partition and the induced vector of bin averages.

2.2. Bin Averages

For any nonnegative f with $\operatorname{supp}(f) \subseteq (-\frac{1}{4}, \frac{1}{4})$ and $\int f = 1$, define the *bin averages*:

$$a_j \;=\; 4n \int_{I_j} f(x)\,dx \;\geq\; 0, \qquad j = 0, \ldots, d-1.$$

The factor $4n$ normalizes so that each a_j equals the average value of f on the bin I_j (since $|I_j| = \frac{1}{4n}$). The constraint $\int f = 1$ becomes

$$\sum_{j=0}^{d-1} a_j \;=\; 4n.$$

Definition 2.2. The *continuous simplex* is

$$A_n \;=\; \left\{ a \in \mathbb{R}^d_{\geq 0} \;\middle|\; \sum_{j=0}^{d-1} a_j = 4n \right\}.$$

Every nonnegative function f with $\int f = 1$ maps to a point in A_n, but not every point in A_n arises from a function (the map loses information about the shape of f within each bin).

2.3. The Lower Bound on $\|f * f\|_\infty$

Remarkably, the entire reduction rests on two elementary observations — one geometric, one analytic.

The next lemma is the basic analytic reduction of Cloninger and Steinerberger [CS17, Lemma 1].

Lemma 2.3 (Reduction to Bin Averages)**.** *For any $f \in \mathcal{F}$ with $\int f = 1$ and corresponding bin averages $a = (a_0, \ldots, a_{d-1})$:*

$$\|f * f\|_{L^\infty} \;\geq\; \max_{\substack{2 \leq \ell \leq 2d \\ k}} \frac{1}{4n\ell} \sum_{\substack{0 \leq i,j \leq d-1 \\ k \leq i+j \leq k+\ell-2}} a_i\, a_j.$$

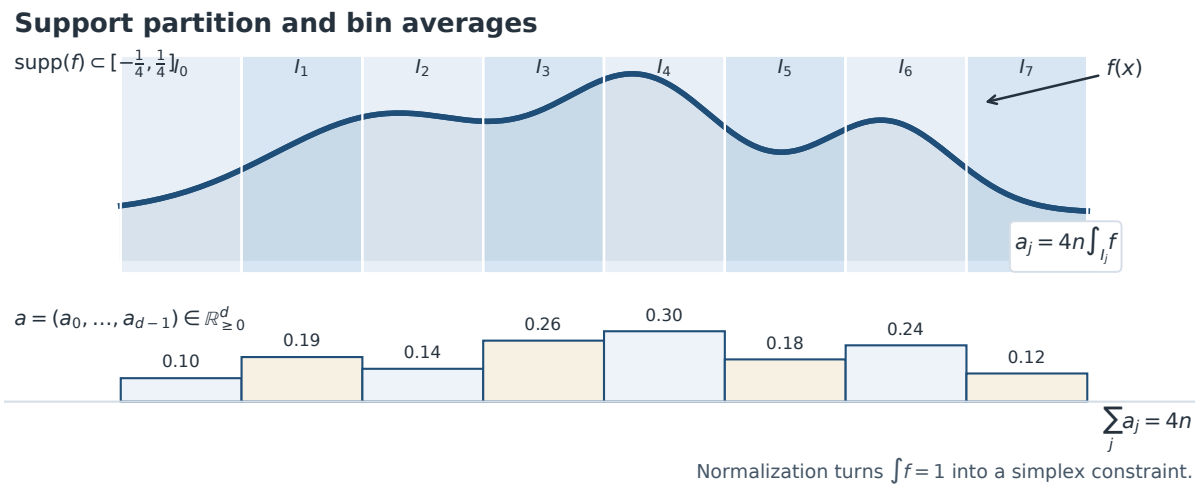

Figure 1: A stylized nonnegative profile supported on $[-\frac{1}{4}, \frac{1}{4}]$, together with the equal-bin partition and the resulting vector of bin averages. The lower panel emphasizes the simplex constraint $\sum_j a_j = 4n$ after normalizing $\int f = 1$.

Proof sketch. **Ingredient 1: Support of cross-convolutions.** Let $f_i = f \cdot \mathbf{1}_{I_i}$ be the restriction of f to bin I_i. Then $f = \sum_i f_i$ and

$$(f * f)(t) \;=\; \sum_{i,j} (f_i * f_j)(t).$$

The convolution $f_i * f_j$ is supported on the Minkowski sum $I_i + I_j$, which is an interval of width $\frac{1}{2n}$. The key point: *the index $i + j$ determines which part of the convolution domain receives the contribution from $f_i * f_j$.*

Ingredient 2: Pigeonhole. For any nonnegative function h and interval J: $\|h\|_{L^\infty} \geq \frac{1}{|J|} \int_J h$.

Combining. Consider a "window" covering sum-indices $s \in \{k, k+1, \ldots, k+\ell-2\}$. The Minkowski sums for consecutive sum-indices overlap (their centers are spaced $\frac{1}{4n}$ apart, each of width $\frac{1}{2n}$), so their union forms an interval of length exactly $\frac{\ell}{4n}$ in the convolution domain. All cross-terms $f_i * f_j$ with $k \leq i + j \leq k + \ell - 2$ have support within this interval, contributing total L^1 mass $\sum \frac{a_i}{4n} \cdot \frac{a_j}{4n}$. Applying the pigeonhole bound:

$$\|f * f\|_{L^\infty} \;\geq\; \frac{1}{\ell/(4n)} \sum_{k \leq i+j \leq k+\ell-2} \frac{a_i\, a_j}{(4n)^2} \;=\; \frac{1}{4n\ell} \sum_{k \leq i+j \leq k+\ell-2} a_i\, a_j.$$

Taking the maximum over all windows gives the result. □

Remark 2.4 (Formal version of the reduction)**.** The Lean file `Sidon/TestValueBounds.lean` proves a direct theorem of this form, called `continuous_test_value_le_ratio`. One extra hypothesis is made explicit there: $\|f * f\|_{L^\infty} < \infty$. This is a bookkeeping issue forced by the use of `ENNReal.toReal` in the formal definition of the ratio; if $\|f * f\|_{L^\infty} = \infty$, then any finite lower bound is automatic anyway.

Remark 2.5 (Intuition)**.** Think of the bin averages a_i as describing "how much mass f puts in each bin." The products $a_i a_j$ measure how much the mass in bins i and j contribute to the autoconvolution. The window argument says: if too much product-mass falls in any fixed interval of the convolution domain, then $f * f$ must be large somewhere in that interval.

The reader may wonder: why not just take the narrowest possible window ($\ell = 2$) everywhere? Because narrow windows see fewer cross-terms, so their sums can be small even when mass is spread across

many bins. The interplay between window width and the number of contributing pairs is what makes the optimization nontrivial.

3. Discretization of the Simplex

Having reduced the problem to minimizing over the continuous simplex A_n, we face a new obstacle: A_n is uncountable. To bring the problem within reach of a computer, we renormalize to the bin-mass coordinates

$$\mu_j = \frac{a_j}{4n}, \qquad \sum_{j=0}^{d-1} \mu_j = 1,$$

and approximate this normalized simplex by a finite lattice. Regular simplex grids are a standard device in finite-dimensional global optimization [BGY14]; here the same geometry is adapted to the composition coordinates used by the branch-and-prune computation.

3.1. The Lattice

In these normalized coordinates, the natural grid has spacing $1/m$.

Definition 3.1. Fix a positive integer m (the "grid resolution"). The *discrete lattice* is

$$B_{n,m} = \left\{ b \in \left(\tfrac{1}{m}\mathbb{N}\right)^d \;\middle|\; \sum_{j=0}^{d-1} b_j = 1 \right\},$$

where each b_j is a nonnegative multiple of $\frac{1}{m}$.

Equivalently, writing $c_j = m \cdot b_j$, the lattice consists of nonnegative integer vectors $c = (c_0, \ldots, c_{d-1})$ with $\sum c_j = m$. These are the *compositions* of the integer m into d nonnegative parts.

Remark 3.2 (Implementation convention)**.** The implementation stores the integer composition c itself. The corresponding bin-average vector is

$$a_j = \frac{4n}{m} c_j.$$

Thus the discretization lemmas are most naturally phrased in the normalized mass coordinates $b_j = c_j/m$, while the test-value formulas remain most natural in the bin-average coordinates a_j.

Figure 2 shows this discretization on a two-dimensional toy simplex, where a continuous point is rounded to a nearby lattice point.

The lattice is useful only if every continuous point has a nearby lattice point — otherwise gaps in the lattice could hide the minimizer.

The relevant approximation statement is Cloninger–Steinerberger [CS17, Lemma 2].

Lemma 3.3 (Lattice as an Approximation)**.** *$B_{n,m}$ is a $\frac{1}{m}$-net of the normalized mass simplex. That is, for every $\mu \in \mathbb{R}^d_{\geq 0}$ with $\sum \mu_j = 1$, there exists $b \in B_{n,m}$ with $\|\mu - b\|_\infty \leq \frac{1}{m}$.*

Proof sketch. Given μ with $\sum \mu_j = 1$, round each coordinate to the nearest multiple of $\frac{1}{m}$, adjusting to maintain the sum constraint. Specifically, use a greedy rounding: process coordinates left to right, maintaining a running deficit $t_i = \sum_{j<i}(b_j - \mu_j)$. Round down first, then alternate to keep $-\frac{1}{m} < t_{i+1} \leq 0$. The last coordinate is set to enforce the exact sum. Each coordinate satisfies $|\mu_j - b_j| \leq \frac{1}{m}$. □

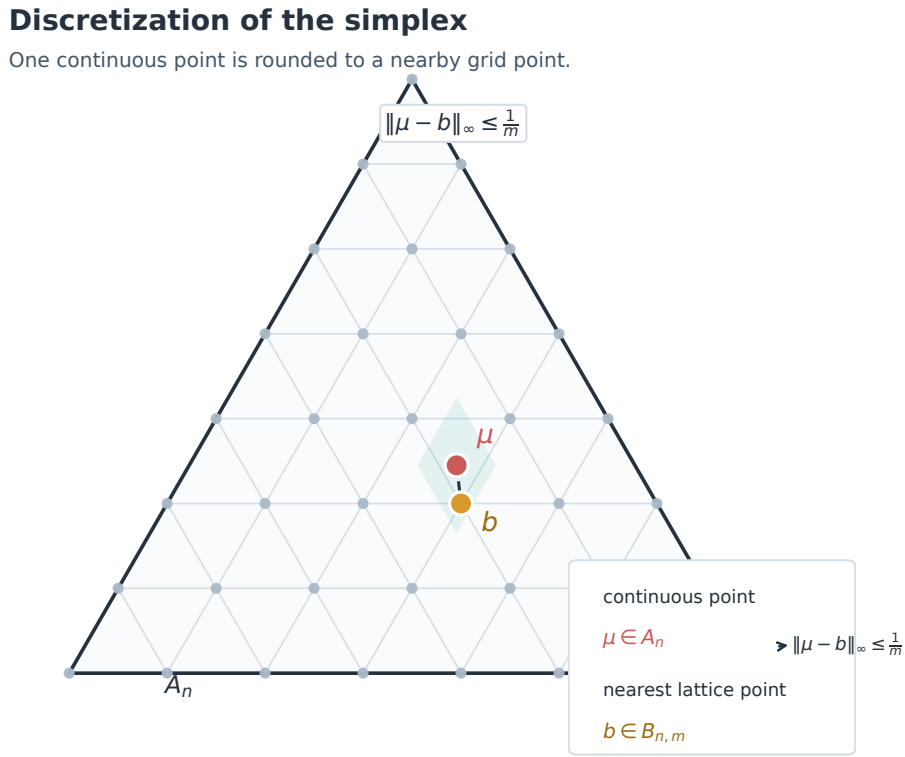

Figure 2: A three-coordinate slice of the normalized simplex. A point μ is rounded to a nearby lattice point $b \in B_{n,m}$, illustrating the ℓ_∞-net statement used in the discretization step.

3.2. The Discretization Error Bound

Since we approximate the normalized mass vector μ by b with per-coordinate error at most $\frac{1}{m}$, the windowed test value changes by a bounded amount.

The next inequality is the corrected discretization estimate replacing the original Lemma 3 in [CS17].

Lemma 3.4 (Discretization Error)**.** *Let $a \in A_n$, write $\mu_i = a_i/(4n)$, and let $b \in B_{n,m}$ satisfy $\|\mu - b\|_\infty \le \frac{1}{m}$. Then, for any admissible window (ℓ, k),*

$$W_{\ell,k}(\mu) \;\ge\; W_{\ell,k}(b) \;-\; \frac{4n}{\ell}\left(\frac{2}{m} + \frac{1}{m^2}\right),$$

where

$$W_{\ell,k}(x) \;=\; \frac{4n}{\ell} \sum_{\substack{0 \le i,j \le d-1 \\ k \le i+j \le k+\ell-2}} x_i x_j .$$

Consequently, if $\tilde{b}_i = 4n\, b_i$ is the associated bin-average vector, then

$$\mathrm{tv}(a) \;\ge\; \mathrm{tv}(\tilde{b}) \;-\; 2n\left(\frac{2}{m} + \frac{1}{m^2}\right).$$

Proof sketch. This is the corrected CS17 discretization bound. For a fixed window, write $\mu = b + \varepsilon$. The perturbation contributes two error terms: a cross term of size $\frac{4n}{\ell} \cdot \frac{2}{m} W$, where $W \le 1$ is the total contributing

mass in that window, and a quadratic term of size $\frac{4n}{\ell} \cdot \frac{1}{m^2}$. Summing them yields the per-window correction

$$\frac{4n}{\ell}\left(\frac{2}{m} + \frac{1}{m^2}\right).$$

Since $\ell \geq 2$, we have $\frac{4n}{\ell} \leq 2n$, which gives the stated global bound for tv. □

Remark 3.5 (The correction term in practice)**.** The raw CS17 correction is $\frac{2}{m} + \frac{1}{m^2}$, but the global level-$n$ correction is $2n\left(\frac{2}{m} + \frac{1}{m^2}\right)$. At the starting level $n = 2$ and $m = 20$, this is 0.41. In the implementation accompanying this paper, this global quantity is used only for coarse caps and reporting; the actual pruning kernel uses the sharper window-dependent threshold described next.

3.3. The Refined (Dynamic) Correction

The global correction bound from Lemma 3.4 is pessimistic: it assumes the worst-case window length and replaces the actual contributing mass by its trivial upper bound. A tighter, position-dependent bound exploits the fact that narrow or edge-positioned windows involve fewer bins.

The window-specific threshold below is the one exploited by Cloninger and Steinerberger [CS17, Eq. (1)] and by the present cascade implementation.

Lemma 3.6 (Dynamic Correction)**.** *For a specific window at position k with length ℓ, let $W_{int} = \sum_{i \in \mathcal{B}(k,\ell)} c_i$ be the total integer mass in contributing bins, where $\mathcal{B}(k,\ell) = \{i : \exists\, j \text{ with } k \leq i + j \leq k + \ell - 2\}$. Then the* window-specific effective threshold *is:*

$$T_{\ell,k} \;=\; c_{\text{target}} + \frac{4n}{\ell} \cdot \frac{1 + 2\,W_{\text{int}}}{m^2}$$

rather than the coarse global threshold $c_{\text{target}} + 2n\left(\frac{2}{m} + \frac{1}{m^2}\right)$.

Proof. Let ws be the integer window sum of the autoconvolution of a composition c with $\sum_i c_i = m$. Since the implementation uses the rescaling $a_i = \frac{4n}{m} c_i$, the corresponding test value is

$$W_{\ell,k}(a) \;=\; \frac{4n}{m^2 \ell}\,\text{ws}.$$

Therefore the corrected dynamic pruning condition is equivalent to

$$\text{int_thresh} \;=\; \left\lfloor \left(c_{\text{target}} \cdot m^2 \cdot \frac{\ell}{4n} + 1 + 2\,W_{\text{int}} + 10^{-9} m^2\right)(1 - 4\epsilon_{\text{mach}}) \right\rfloor,$$

which is the corrected form now used in `run_cascade.py`. The additive $10^{-9}m^2$ term raises the threshold, while the factor $(1 - 4\epsilon_{\text{mach}})$ lowers it slightly before flooring; the additive margin dominates, so the net effect is conservative. □

Remark 3.7 (Why dynamic is better)**.** The dynamic correction keeps track of the actual contributing mass W_{int} and retains the favorable factor $\frac{4n}{\ell}$. Long windows and edge windows therefore carry smaller corrections than the worst-case global bound, which is why the cascade can prune far more aggressively than a uniform-threshold search.

4. The Windowed Test Value

With the discretization machinery in hand, we can describe precisely what the algorithm computes at each lattice point.

4.1. Definition

Definition 4.1. For a mass vector $a = (a_0, \ldots, a_{d-1})$ with $a_i \geq 0$ and $\sum a_i = 4n$, the *autoconvolution* is the vector of length $2d - 1$:

$$\mathrm{conv}[s] \;=\; \sum_{\substack{i+j=s \\ 0\leq i,j\leq d-1}} a_i\, a_j, \qquad s = 0, 1, \ldots, 2(d-1).$$

The *test value* is

$$\mathrm{tv}(a) \;=\; \max_{\substack{2\leq \ell\leq 2d \\ 0\leq k\leq 2(d-1)-(\ell-2)}} \frac{1}{4n\ell} \sum_{s=k}^{k+\ell-2} \mathrm{conv}[s].$$

In words: compute the autoconvolution, slide a window of each possible width ℓ across it, sum the $\ell - 1$ consecutive entries in each window position, normalize by $4n\ell$, and take the maximum.

Remark 4.2 (Integer implementation coordinates)**.** The code stores compositions $c = (c_0, \ldots, c_{d-1}) \in \mathbb{N}^d$ with $\sum_i c_i = m$ and recovers the bin-average vector by

$$a_i \;=\; \frac{4n}{m}\, c_i.$$

All pruning formulas below are stated in these implementation coordinates.

4.2. Efficient Computation via Prefix Sums

The test value can be computed efficiently using prefix sums. Let $P[s] = \sum_{k=0}^{s} \mathrm{conv}[k]$ be the prefix sum of the autoconvolution. Then the windowed sum from s_{lo} to s_{hi} is:

$$\sum_{s=s_{\mathrm{lo}}}^{s_{\mathrm{hi}}} \mathrm{conv}[s] \;=\; P[s_{\mathrm{hi}}] - P[s_{\mathrm{lo}} - 1] \qquad (\text{with } P[-1] = 0).$$

Remark 4.3 (Complexity)**.** Computing the autoconvolution takes $O(d^2)$ time. An FFT-based method would give $O(d \log d)$, but for the small d values used here the direct method is faster in practice. The prefix sum takes $O(d)$, and the complete window scan takes another $O(d^2)$. Thus the overall cost per lattice point is $O(d^2)$.

4.3. Reversal Symmetry

A key structural observation cuts the search space in half. If we reverse the order of the bins, the autoconvolution peak does not change — intuitively, because convolution does not care which direction we label the support.

Theorem 4.4 (Autoconvolution Reversal Symmetry)**.** *For any mass vector a,* $\mathrm{tv}(a) = \mathrm{tv}(\mathrm{rev}(a))$*, where* $\mathrm{rev}(a) = (a_{d-1}, \ldots, a_0)$.

Proof. The argument has three moving parts.

Step 1: Autoconvolution reversal. Let $a' = \mathrm{rev}(a)$, so $a'_i = a_{d-1-i}$. Then:

$$\mathrm{conv}'[s] \;=\; \sum_{i+j=s} a'_i a'_j \;=\; \sum_{i+j=s} a_{d-1-i}\, a_{d-1-j}.$$

Substituting $i' = d - 1 - i$, $j' = d - 1 - j$ (so $i' + j' = 2(d-1) - s$):

$$\mathrm{conv}'[s] \;=\; \mathrm{conv}[2(d-1) - s].$$

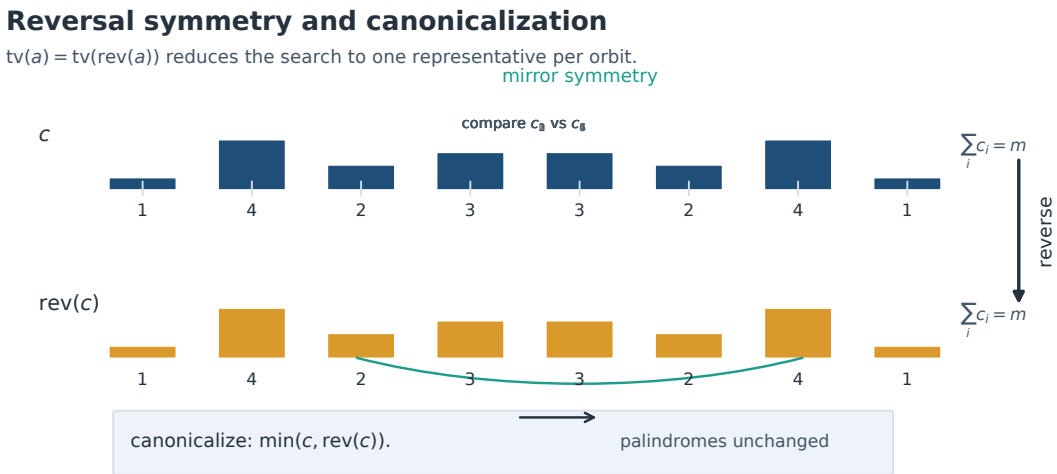

Figure 3: Reversal symmetry of the test value and the canonicalization rule $c \mapsto \min(c, \mathrm{rev}(c))$. Palindromic compositions are fixed by the involution, while non-palindromic pairs contribute only one canonical representative to the search.

That is, the autoconvolution of the reversal is the reversal of the autoconvolution.

Step 2: Window-sum bijection. A window sum of width $w = \ell - 1$ starting at s_0 transforms under reversal to a window sum of the same width starting at $2(d-1) - s_0 - w + 1$:

$$\begin{aligned}\sum_{s=s_0}^{s_0+w-1} \mathrm{conv}'[s] &= \sum_{s=s_0}^{s_0+w-1} \mathrm{conv}[2(d-1) - s] \\ &= \sum_{s'=2(d-1)-s_0-w+1}^{2(d-1)-s_0} \mathrm{conv}[s'].\end{aligned}$$

This is a window of the same width w, so the multiset of all window sums at width w is identical for conv and conv′.

Step 3: Normalization preserves the maximum. The normalization $\frac{1}{4n\ell}$ depends only on ℓ, not the window position. Since the multiset of window sums is identical at each width, $\mathrm{tv}(a) = \mathrm{tv}(a')$. □

Corollary 4.5 (Canonical Filter)**.** *It suffices to check only* canonical *integer compositions c with $c \le \mathrm{rev}(c)$ lexicographically. The associated bin-average vectors have the same test value, so this halves the search space (up to palindromes).*

Figure 3 depicts the reversal orbit of a composition and the lexicographic canonicalization used to choose one representative from each non-palindromic pair.

Even after exploiting reversal symmetry, the number of implementation states to check — roughly $\binom{m+d-1}{d-1}/2$ — grows rapidly with d and m. To keep the computation tractable, we need ways to rule out large swathes of the simplex without computing the full test value.

5. Pruning Strategies

5.1. Asymmetry Pruning

At first glance, the following bound might seem too crude to be useful: it ignores everything about f except how much mass sits on each half of the support. Yet in practice, this single test eliminates the vast majority of configurations at every cascade level.

Theorem 5.1 (Asymmetry Bound)**.** *Let $f \geq 0$ with $\mathrm{supp}(f) \subseteq (-\frac{1}{4}, \frac{1}{4})$ and $\int f = 1$. Let $L = \int_{-1/4}^{0} f$ be the left-half mass. Then:*

$$\|f * f\|_{L^\infty} \;\geq\; 2L^2.$$

Proof. Consider the left-left convolution:

$$(f * f)(t) \;=\; \int f(x)\, f(t-x)\, dx \;\geq\; \int_{-1/4}^{0} f(x)\, f(t-x)\, \mathbf{1}_{[-1/4,0]}(t-x)\, dx$$

for $t \in (-\frac{1}{2}, 0)$. By Fubini's theorem, the total mass of $f * f$ restricted to $(-\frac{1}{2}, 0)$ is at least L^2 (the left-left contribution). Since $|(-\frac{1}{2}, 0)| = \frac{1}{2}$:

$$\|f * f\|_{L^\infty} \;\geq\; \frac{L^2}{1/2} \;=\; 2L^2.$$

By symmetry, the same holds for the right-half mass $R = 1 - L$: $\|f * f\|_\infty \geq 2R^2$. Therefore $\|f * f\|_\infty \geq 2\max(L^2, R^2)$. □

Corollary 5.2 (Asymmetry Threshold)**.** *If $2L^2 \geq c_{\text{target}}$, i.e., $L \geq \sqrt{c_{\text{target}}/2}$, then the configuration is ruled out. The threshold left-mass fraction is:*

$$p^* \;=\; \sqrt{c_{\text{target}}/2}.$$

For $c_{\text{target}} = 1.4$: $p^* \approx 0.8367$.

Remark 5.3 (Exactness for step functions and under refinement)**.** For a step function already represented by a composition $c = (c_0, \ldots, c_{d-1})$ with $\sum c_i = m$, the left-half mass is

$$L \;=\; \sum_{i<n} \frac{c_i}{m},$$

because the midpoint $x = 0$ lies exactly on a bin boundary. This quantity is also preserved exactly under dyadic refinement: the left child bins sum to the left parent bins. Thus asymmetry pruning is an exact statement for discrete step functions inside the cascade.

Remark 5.4 (Continuous-transfer caveat)**.** The asymmetry bound is a *direct* L^∞ bound on $f * f$. It does *not* go through the discretization framework of Lemma 3.4. Accordingly, one must distinguish two roles for asymmetry:

- for an *exact step function* represented by a composition c, the criterion is immediate and needs no correction term;
- inside the certificate computation, it serves as an inexpensive early pruning rule before the more expensive window scan.

The final Lean theorem no longer depends on carrying asymmetry certificates through the continuous-to-discrete transfer. Instead, the formal theorem imports a stronger terminal-grid window certificate at $d = 128$, so asymmetry remains part of the trusted external computation but not part of the final analytic reduction.

Remark 5.5 (What the certificate computation prunes by)**.** Earlier drafts experimented with additional block-sum and two-peak heuristics. The recorded cascade does not rely on them. The certificate computation prunes by asymmetry, the corrected window-dependent dynamic threshold, reversal canonicalization, and the single-bin cap used during child generation. In the Lean development, the external computation may use all of these rules, but the imported certificate ultimately records only a terminal window witness.

5.2. Canonical Symmetry Exploitation

If c is an implementation-space composition and $\tilde{c}_i = \frac{4n}{m} c_i$ is its associated bin-average vector, then Theorem 4.4 gives $\mathrm{tv}(\tilde{c}) = \mathrm{tv}(\widetilde{\mathrm{rev}(c)})$. This motivates the following notion of canonicality.

Definition 5.6. A composition c is *canonical* if $c \le \mathrm{rev}(c)$ lexicographically. That is, comparing c_0 with c_{d-1}, then c_1 with c_{d-2}, etc., c comes first (or they are equal, i.e., c is a palindrome).

Proposition 5.7 (Canonical Count)**.** *For integer compositions of a fixed total mass M into d parts, the number of canonical compositions is $\frac{N+P}{2}$, where $N = \binom{M+d-1}{d-1}$ is the total count and P is the number of palindromes.*

Proof. Every composition falls into exactly one of three categories: palindromes (those equal to their own reversal), strictly canonical non-palindromes ($b < \mathrm{rev}(b)$), and their reversals. Since reversal is an involution with no fixed points outside the palindromes, the second and third categories have equal size C. Counting gives $N = P + 2C$, so the canonical count is $P + C = (N + P)/2$. □

Remark 5.8 (Loop-bound tightening for $d = 4$)**.** For $b = (c_0, c_1, c_2, c_3)$ with total mass M and $c_3 = M - c_0 - c_1 - c_2$, canonicality ($b \le \mathrm{rev}(b)$) requires $c_0 \le c_3$, giving three loop bounds:

- $c_0 \le \lfloor M/2 \rfloor$ (if $c_0 > M/2$ then $c_3 < c_0$).
- $c_1 \le M - 2c_0$ (need $c_2 \ge 0$ with $c_1 + c_2 \le M - 2c_0$).
- $c_2 \le M - 2c_0 - c_1$ (ensures $c_3 \ge c_0$).

Plus a palindrome filter: when $c_0 = c_3$, require $c_1 \le c_2$.

For $d = 6$, analogous bounds apply: $c_0 \le \lfloor M/2 \rfloor$, $c_4 \le M - c_0 - c_1 - c_2 - c_3 - c_0$, with equality checks for the inner pairs when outer pairs tie.

6. The Multiscale Cascade

Brute-force enumeration at fine resolution is hopelessly infeasible — even modest parameters yield billions of lattice points. The key idea is to start coarse: most configurations can be ruled out cheaply at low resolution, and only the stubborn survivors need to be examined more closely.

6.1. Overview

Step 1: Start coarse. At level L_0, enumerate all canonical compositions of m into $d_0 = 2n_0$ parts (e.g., $n_0 = 2$, $d_0 = 4$).

Step 2: Prune. For each composition, apply asymmetry pruning and then scan windows using the corrected dynamic threshold. If some window certifies the target bound, the composition is eliminated.

Step 3: Refine survivors. Each surviving composition at resolution d is refined to resolution $2d$ by *dyadic splitting*: each bin is split into two sub-bins.

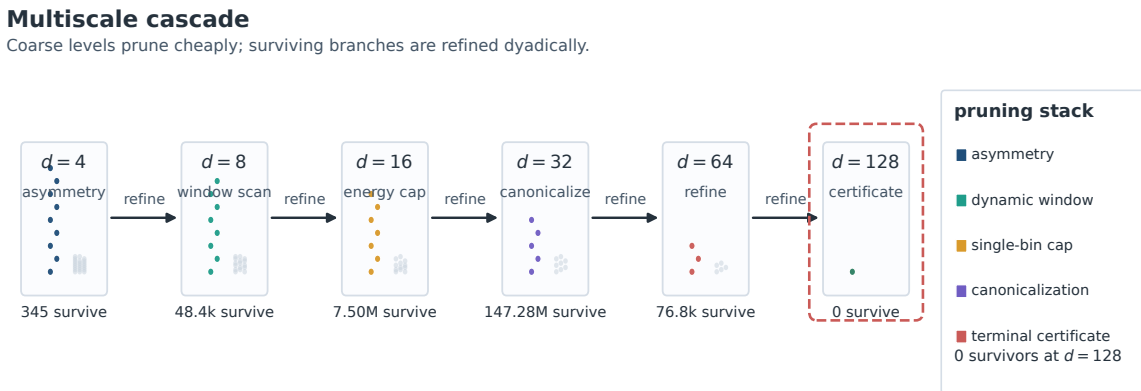

Figure 4: Stylized overview of the multiscale cascade. The annotations record the exact survivor counts from the terminal certificate run: 345 at $d = 4$, then 48,443, 7,499,382, 147,279,894, 76,829, and finally 0 at $d = 128$.

Step 4: Repeat. Prune the refined compositions. Continue until either all are eliminated (proof complete) or a computational budget is reached.

Figure 4 summarizes the recorded certificate run `data/cpu_cascade_20260319_201644.json`, including the exact survivor counts at each dyadic level.

6.2. Dyadic Splitting (Refinement)

Given a parent composition $b = (b_0, \ldots, b_{d-1})$ at resolution d, the refinement produces children $c = (c_0, c_1, \ldots, c_{2d-1})$ at resolution $2d$, where:

$$c_{2i} + c_{2i+1} \;=\; b_i \qquad \text{for each } i = 0, \ldots, d-1,$$

with $c_{2i}, c_{2i+1} \geq 0$ and both being nonnegative integers.

6.3. Energy Cap

One might worry that generating all possible children is wasteful. Indeed, many children are immediately ruled out by the single-bin diagonal window. The energy cap avoids generating them in the first place.

Recall that $a_j = \frac{4n_{\text{child}}}{m} c_j$ and the $\ell = 2$ diagonal window at position $2j$ gives

$$\mathrm{tv}(a) \;\geq\; d_{\text{child}} \frac{c_j^2}{m^2}, \qquad d_{\text{child}} = 2n_{\text{child}}.$$

This yields the test-value cap

$$x_{\text{cap}}^{\text{TV}} \;=\; \left\lfloor m \sqrt{\frac{c_{\text{target}} + 2n_{\text{child}}\left(\frac{2}{m} + \frac{1}{m^2}\right) + 10^{-9}}{d_{\text{child}}}} \right\rfloor.$$

The implementation also applies the sharper direct cap

$$x_{\text{cap}}^{\text{CS}} \;=\; \left\lfloor m \sqrt{\frac{c_{\text{target}}}{d_{\text{child}}}} \right\rfloor,$$

and uses $x_{\text{cap}} = \min\left(x_{\text{cap}}^{\text{TV}}, x_{\text{cap}}^{\text{CS}}\right)$.

Proposition 6.1 (Energy Cap Soundness)**.** *If a child has some bin $c_j > x_{\text{cap}}^{\text{CS}}$, then any nonnegative function with those child-bin masses already satisfies $\|f * f\|_{L^\infty} > c_{\text{target}}$.*

Proof. Let $M_j = c_j/m$ be the mass in the jth child bin. Restricting f to that single bin produces a nonnegative function f_j supported on an interval of length $1/d_{\text{child}}$ and with total mass M_j. By the same averaging argument used for asymmetry pruning,

$$\|f * f\|_{L^\infty} \;\geq\; \|f_j * f_j\|_{L^\infty} \;\geq\; \frac{M_j^2}{1/d_{\text{child}}} \;=\; d_{\text{child}}\,\frac{c_j^2}{m^2}.$$

If $c_j > m\sqrt{c_{\text{target}}/d_{\text{child}}}$, the right-hand side exceeds c_{target}, which proves the claim. □

Remark 6.2 (Tighter Cauchy–Schwarz cap)**.** The looser cap $x_{\text{cap}}^{\text{TV}}$ mirrors the older global correction narrative, while the direct cap $x_{\text{cap}}^{\text{CS}}$ is the effective one in the implementation. It is tighter and does not need any discretization correction term.

6.4. Terminal Certificate

Theorem 6.3 (Recorded Terminal-Grid Certificate)**.** *For every composition $c = (c_0, \ldots, c_{127}) \in \mathbb{N}^{128}$ with $\sum_{i=0}^{127} c_i = 20$, let*

$$a_i \;=\; \frac{4 \cdot 64}{20}\, c_i, \qquad i = 0, \ldots, 127,$$

and let conv_a be the discrete autoconvolution of a. Then there exist integers $\ell \geq 2$ and s_{lo} such that

$$\frac{1}{4 \cdot 64 \cdot \ell} \sum_{s=s_{\text{lo}}}^{s_{\text{lo}}+\ell-2} \text{conv}_a[s] \;>\; \frac{7}{5} + \frac{4 \cdot 64}{\ell}\left(\frac{1}{20^2} + \frac{2}{20^2} \sum_{i \in \mathcal{B}(64,\ell,s_{\text{lo}})} c_i\right),$$

where $\mathcal{B}(64, \ell, s_{\text{lo}})$ is the set of contributing bins for that window.

Status. This is the finite computational statement imported into Lean as the axiom `cascade_all_pruned` in `Sidon/FinalResult.lean`. It is extracted from the recorded certificate run `data/cpu_cascade_20260319_201644.json`, whose cascade starts at $d = 4$, refines dyadically to $d = 128$, and terminates with zero survivors. The purpose of the algorithmic discussion in this section is to explain how that external computation is organized and why the resulting certificate is credible. □

6.5. Canonical Handling Across Levels

At level L_0, only canonical compositions are generated. At levels L_1 and beyond, the situation is more subtle:

- Children of a canonical parent P are *not* necessarily canonical.
- The reversal $\text{rev}(P)$ is *not* in the parent list (unless P is a palindrome), so its children are not generated directly.
- However, if C is a child of P, then $\text{rev}(C)$ is a child of $\text{rev}(P)$. Since $\text{tv}(C) = \text{tv}(\text{rev}(C))$ (Theorem 4.4), testing C suffices for both.

Therefore at levels ≥ 1:

1. Generate *all* children of each parent (no canonical filter during generation).
2. Test each child against the dynamic threshold.
3. *Canonicalize* survivors: replace each with $\min(C, \text{rev}(C))$ lexicographically.

4. *Deduplicate*: remove duplicate canonical forms (children from different parents, or a palindrome parent producing both C and $\mathrm{rev}(C)$).

Proposition 6.4 (Completeness of Canonical Handling)**.** *This procedure tests every composition at the child resolution: if C' is any child-level composition and $P' = parent(C')$, then either P' or $\mathrm{rev}(P')$ is in the parent list, and C' or $\mathrm{rev}(C')$ is among the tested children.*

6.6. Fused Generate-and-Prune Kernel

At deep cascade levels, each parent may have millions of children. The fused kernel generates and tests each child inline via a Cartesian-product odometer, never materializing the full children array:

- **Incremental autoconvolution:** when only the last cursor position changes, the raw autoconvolution is updated in $O(d)$ instead of $O(d^2)$.
- **Quick-check:** re-tries the previous killing window before a full scan.
- **Subtree pruning:** on deep carries, checks the partial autoconvolution of fixed bins against worst-case unfixed mass; if it exceeds the threshold, the entire subtree is skipped.

7. The Complete Proof

7.1. Algorithm

Algorithm 1 Cascade Search For Terminal Coverage

Require: c_{target}, grid resolution m, starting half-bins n_0, max levels $L_{\max}$

```
d_0 ← 2n_0
Generate all canonical compositions of m into d_0 parts          ▷ implementation coordinates
Apply asymmetry pruning (threshold √(c_target/2))
For every remaining composition, scan all windows and prune if some window exceeds the dynamic
  threshold from Lemma 3.6
survivors_0 ← unpruned compositions
for k = 1, 2, ..., L_max do
    d_k ← 2d_{k−1};   n_k ← 2n_{k−1}
    Compute the single-bin child cap x_cap = min(x_cap^TV, x_cap^CS)
    for all parent P ∈ survivors_{k−1} do
        for all children C of P (dyadic split with energy cap) do
            Apply asymmetry pruning to C
            if not asymmetry-pruned then
                Scan all windows of C via prefix-sum autoconvolution
                if some window exceeds the dynamic threshold then
                    Prune C
                else
                    C survives; canonicalize and store
                end if
            end if
        end for
    end for
    Deduplicate survivors (canonicalize → sort → scan)
    survivors_k ← deduplicated survivors
    if |survivors_k| = 0 then
        return ZERO-SURVIVORS
    end if
end for
return INCONCLUSIVE: |survivors_{L_max}| compositions remain
```

7.2. Main Theorem

Theorem 7.1 (Formal Main Result)**.** *Let $f : \mathbb{R} \to \mathbb{R}$ be nonnegative, satisfy* $\operatorname{supp}(f) \subseteq (-\frac{1}{4}, \frac{1}{4})$*, and obey* $\int f > 0$*. If $\|f * f\|_{L^\infty} < \infty$, then*

$$\frac{\|f * f\|_{L^\infty}}{\left(\int f\right)^2} \geq \frac{7}{5}.$$

Consequently $C_{1a} \geq \frac{7}{5} = 1.4$.

Proof. If $\|f * f\|_{L^\infty} = \infty$, then the displayed inequality is automatic, so only the finite case needs work.

Normalize

$$g = \frac{f}{\int f},$$

so that $\int g = 1$. The autoconvolution ratio is scale-invariant, so it suffices to prove $\|g * g\|_{L^\infty} \geq \frac{7}{5}$.

Discretize g using the cumulative-floor rule at parameters $n = 64$ and $m = 20$, obtaining a composition $c \in \mathbb{N}^{128}$ with $\sum_i c_i = 20$. By Theorem 6.3, this composition admits a window (ℓ, s_{lo}) whose test value exceeds the dynamic threshold at target $\frac{7}{5}$. Lemma 3.6 is precisely the statement that such a window certificate forces $\|g * g\|_{L^\infty} \geq \frac{7}{5}$. By scale invariance, the same lower bound holds for the original f.

For the constant C_{1a}, functions with $\|f * f\|_{L^\infty} = \infty$ do not affect the infimum, so the displayed bound implies $C_{1a} \geq \frac{7}{5}$. □

8. Correctness Invariants

Stating the algorithm is only half the battle. For the proof to carry mathematical weight, we must verify that the implementation faithfully realizes the specification. Four invariants pin down exactly what "faithful" means. In the accompanying Lean development, these invariants justify trust in the external computation behind Theorem 6.3; the analytic transfer from that terminal certificate to the continuous theorem is formalized separately.

8.1. Invariant 1: Complete Enumeration

Every implementation-space composition $c \in \mathbb{N}^d$ with $\sum_i c_i = m$ must be either directly checked or shown to be covered by a checked ancestor and all its refinements.

- At L_0: all $\binom{m+d_0-1}{d_0-1}$ compositions are enumerated (in canonical form, with reversal symmetry). Verified by comparing the generator output count against the stars-and-bars formula, and by checking completeness, uniqueness, non-negativity, and sum constraints.
- At L_k ($k \geq 1$): all children of all surviving parents are generated (no sampling or probabilistic skipping).

8.2. Invariant 2: All Windows Checked

For each composition that is not eliminated by asymmetry, every valid window (ℓ, k) with $2 \leq \ell \leq 2d$ and all valid starting positions k must be tested before the composition is declared a survivor. Early exit occurs only *after* a pruning window is found — never for survivors.

8.3. Invariant 3: Conservative Arithmetic

Floating-point arithmetic is the Achilles' heel of computer-assisted proofs: a single rounding error in the wrong direction could cause a valid configuration to be falsely pruned, invalidating the entire argument. We guard against this as follows:

- Integer coordinates: the autoconvolution is computed in integer arithmetic ($c_i c_j$ are exact integers), avoiding round-off entirely.
- Conservative threshold: the code evaluates the corrected integer threshold

$$c_{\text{target}} m^2 \frac{\ell}{4n} + 1 + 2W_{\text{int}},$$

  together with an added $10^{-9}m^2$ margin and a final $(1 - 4\epsilon_{\text{mach}})$ factor before flooring.
- The multiplicative factor lowers the threshold slightly, but the additive $10^{-9}m^2$ margin is overwhelmingly larger, so the net effect remains conservative: the code may miss a borderline prune, but it does not falsely prune.

8.4. Invariant 4: Refinement Completeness

Every refinement of a surviving parent must be checked at the finer level. The dyadic splitting with energy cap generates all valid children (Section 6.3), and children exceeding the energy cap are provably pruned.

9. Computational Verification

The correctness of the implementation has been verified through multiple independent checks.

9.1. Composition Generators

- **Completeness**: For 7 parameter combinations (d, S), the generator output count matches $\binom{S+d-1}{d-1}$ exactly. All outputs sum to S, are non-negative, and are unique.
- **Specialized ≡ generic**: The $d = 4$ and $d = 6$ specialized generators produce identical sets to the generic stack-based generator.
- **Canonical correctness**: For 8 parameter combinations, canonical generators produce exactly $\frac{N+P}{2}$ outputs, all satisfying $b \leq \mathrm{rev}(b)$, with every non-canonical b having $\mathrm{rev}(b)$ in the output.
- **Batch boundaries**: Across 9 configurations and 8 batch sizes, output is identical regardless of where batch boundaries fall (160 checks, all pass).

9.2. Test Value Computation

- Reversal symmetry ($\mathrm{tv}(b) = \mathrm{tv}(\mathrm{rev}(b))$) verified exhaustively for $d = 4, S = 8$ (165 compositions) and $d = 6, S = 6$ (462 compositions), plus 1,500 random tests. Zero mismatches.
- Pure-Python reference implementation agrees with Numba JIT kernel on all 891 canonical compositions at $d = 4, m = 20$.

9.3. Pruning Correctness

- Window pruning: zero compositions are pruned whose true test value is below the dynamic threshold.
- Window pruning: zero compositions fail to be pruned when their true test value exceeds the dynamic threshold.
- Asymmetry filter: exhaustive step-function checks at coarse levels confirm the expected lower bound on the exact discretized profiles, and the filter is exactly symmetric under reversal.

9.4. MATLAB Baseline Comparison

The cascade prover's dynamic threshold formula has been verified algebraically equivalent to the original MATLAB code of Cloninger–Steinerberger, with the Python implementation being strictly more conservative (harder to prune) due to the explicit $10^{-9}m^2$ safety margin.

9.5. Final Cascade Run

The recorded CPU cascade in `data/cpu_cascade_20260319_201644.json` reaches zero survivors at level L_5, corresponding to dyadic refinement from $d = 4$ through $d = 128$. The companion log `data/cpu_job.log` records the same terminal outcome after roughly 70.08 hours of wall-clock time. In the accompanying Lean formalization, this finite computation is packaged as the axiom `cascade_all_pruned`, which states directly that every terminal composition at $d = 128$ (equivalently, $n = 64$) with total mass $m = 20$ has some window whose test value exceeds the corrected dynamic threshold at target $\frac{7}{5}$.

9.6. Lean Formalization Status

The accompanying formalization consists of a modular Lean 4 development under `lean/Sidon/` and a consolidated export in `lean/complete_proof.lean`. It is carried out in Lean 4 [dMU21], on top of the mathlib

library [The20]. The development formalizes the reduction in separate files for definitions, step functions, continuous test values, discretization error, asymmetry, the Cauchy–Schwarz single-bin bound, refinement support, subtree pruning, and the final normalization argument. In particular, `Sidon/TestValueBounds.lean`, `Sidon/DiscretizationError.lean`, and `Sidon/FinalResult.lean` now carry the analytic argument all the way to the direct lower bound $\frac{7}{5}$.

One important update relative to earlier drafts is that the former informal finiteness step has been replaced by an explicit hypothesis $\|f * f\|_{L^\infty} < \infty$ in the formal theorem chain. This is the clean way to avoid the pathology that `ENNReal.toReal` sends ∞ to 0.

At the same time, the Lean development is not yet completely assumption-free. The final reduction rests on a single explicit imported axiom:

- the computational certificate `cascade_all_pruned`, extracted from the recorded L_5 cascade run at $d = 128$.

Accordingly, the present write-up is phrased around the same direct terminal-grid certificate: the trusted external computation ends at $d = 128$, and the Lean theorem imports exactly that statement.

10. Discussion

10.1. Relationship to the Original CS17 Proof

Our framework is a direct extension of Cloninger–Steinerberger [CS17]. The key differences:

1. **Dynamic correction**: We use a sharper per-window correction rather than only a coarse global bound. This improves pruning while preserving rigor.
2. **Multiscale cascade**: CS17 used $m = 50$ and refined from $n = 3$ through $n = 24$. We use $m = 20$ with dyadic refinement from $d = 4$ through $d = 128$, exploiting the fused kernel's efficiency.
3. **Fused generation**: Instead of materializing all children and then testing, we generate and test each child inline, reducing memory from $O(N_{\text{children}} \cdot d)$ to $O(d)$ per parent.
4. **Integer arithmetic**: Autoconvolution and threshold comparisons use exact integer operations, with floating-point involved only in the conservative threshold computation.

10.2. Limitations

- The proof is *computer-assisted*: the mathematical argument reduces correctness to a finite computation, but that computation must be performed by a machine. The invariants in Section 8 ensure that the computation is trustworthy.
- The approach is limited by the exponential growth of the composition count with d and m. The multiscale cascade mitigates this, but very aggressive targets (c_{target} close to C_{1a}) will require more cascade levels and higher computational cost.
- The Lean theorem imports the terminal certificate `cascade_all_pruned` rather than replaying the full 70-hour search inside the Lean kernel. Eliminating this final axiom would require a kernel-level certification of the same finite computation.
- The formal statement isolates the case $\|f * f\|_{L^\infty} < \infty$ because of the way the ratio is encoded in Lean. This is harmless for the lower bound on C_{1a}, since functions with unbounded self-convolution already satisfy every finite lower bound vacuously.

Acknowledgments

This work builds on the foundational algorithm of Cloninger and Steinerberger [CS17]. The constant C_{1a} is tracked in Tao et al.'s optimization constants repository [Tao].

References

[BGY14] Immanuel M. Bomze, Stefan Gollowitzer, and E. Alper Yildirim, *Rounding on the standard simplex: Regular grids for global optimization*, Journal of Global Optimization **59** (2014), no. 2–3, 243–258.

[CRV10] Javier Cilleruelo, Imre Z. Ruzsa, and Carlos Vinuesa, *Generalized sidon sets*, Advances in Mathematics **225** (2010), no. 5, 2786–2807.

[CS17] Alexander Cloninger and Stefan Steinerberger, *On suprema of autoconvolutions with an application to sidon sets*, Proceedings of the American Mathematical Society **145** (2017), no. 8, 3191–3200.

[dMU21] Leonardo de Moura and Sebastian Ullrich, *The lean 4 theorem prover and programming language*, Automated Deduction – CADE 28, Lecture Notes in Computer Science, vol. 12699, Springer, Cham, 2021, pp. 625–635.

[ET41] P. Erdős and P. Turán, *On a problem of sidon in additive number theory, and on some related problems*, Journal of the London Mathematical Society **s1-16** (1941), no. 4, 212–215.

[MO06] Greg Martin and Kevin O'Bryant, *Constructions of generalized sidon sets*, Journal of Combinatorial Theory, Series A **113** (2006), no. 4, 591–607.

[MO07] ———, *The symmetric subset problem in continuous ramsey theory*, Experimental Mathematics **16** (2007), no. 2, 145–165.

[MO09] ———, *The supremum of autoconvolutions, with applications to additive number theory*, Illinois Journal of Mathematics **53** (2009), no. 1, 219–235.

[MV10] Mate Matolcsi and Carlos Vinuesa, *Improved bounds on the supremum of autoconvolutions*, Journal of Mathematical Analysis and Applications **372** (2010), no. 2, 439–447.

[SS02] Andrzej Schinzel and Wolfgang M. Schmidt, *Comparison of L^1- and L^∞-norms of squares of polynomials*, Acta Arithmetica **104** (2002), no. 3, 283–296.

[Tao] Terence Tao, *An autocorrelation constant related to sidon sets*, https://teorth.github.io/optimizationproblems/constants/1a.html, Accessed 2026-03-26.

[The20] The mathlib Community, *The lean mathematical library*, Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs, 2020, pp. 367–381.